

```

class NQueensBacktracking {

    final int N = 8;

    final int PRINT_LIMIT = 4;

    int solutionCount = 0;

    int printedSolutions = 0;

    boolean isSafe(int board[][], int row, int col) {

        int i, j;

        // Check row on left side
        for (i = 0; i < col; i++)
            if (board[row][i] == 1)
                return false;

        // Check upper diagonal on left side
        for (i = row, j = col; i >= 0 && j >= 0; i--, j--)
            if (board[i][j] == 1)
                return false;

        // Check lower diagonal on left side
        for (i = row, j = col; j >= 0 && i < N; i++, j--)
            if (board[i][j] == 1)
                return false;

        return true;
    }

    void solveNQUtil(int board[][], int col) {

        if (col >= N) {

            solutionCount++;

            if (printedSolutions < PRINT_LIMIT) {

```

```

        System.out.println("Solution #" + solutionCount + ":");
        printSolution(board);
        System.out.println();
        printedSolutions++;
    }
    return;
}

for (int i = 0; i < N; i++) {
    if (isSafe(board, i, col)) {
        board[i][col] = 1;

        solveNQUtil(board, col + 1);

        board[i][col] = 0; // Backtrack
    }
}

}

void solveNQ() {
    int board[][] = new int[N][N];

    solveNQUtil(board, 0);

    if (solutionCount == 0) {
        System.out.println("Solution does not exist");
        return;
    }

    System.out.println("Total solutions: " + solutionCount);
}

```

```

void printSolution(int board[][]) {
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++)
            if (board[i][j] == 1)
                System.out.print("Q ");
            else
                System.out.print("- ");
        System.out.println();
    }
}

public static void main(String args[]) {
    NQueensBacktracking obj = new NQueensBacktracking();
    obj.solveNQ();
}

```

### OUTPUT:-

Solution #1:

```

Q -----
-----Q -
----Q ---
-----Q
-Q -----
---Q ----
-----Q --
--Q -----

```

Solution #2:

Q-----  
-----Q-  
---Q-----  
-----Q--  
-----Q  
-Q-----  
----Q---  
--Q-----

Solution #3:

Q-----  
-----Q--  
-----Q  
--Q-----  
-----Q-  
---Q-----  
-Q-----  
----Q---

Solution #4:

Q-----  
----Q---  
-----Q  
-----Q--  
--Q-----  
-----Q-  
-Q-----  
---Q-----

Total solutions: 92